CS5701(Quantitative Data Analysis)

Lab 1: Part 2

The aim of this part of the lab exercise is to give you independent practical experience in performing descriptive analysis of a data set using R Studio and R Markdown.

Before you start

This independent practice is not assessed, not compulsory, and has no effect on your grade or profile - work through it at your own pace, and ask us (or the GTAs) for help at any point.

By working through the core tasks below, you will have practiced:

- loading a dataset in R and checking it against its documentation,
- producing appropriate numerical summaries for categorical and numerical variables,
- producing appropriate graphical summaries for categorical and numerical variables,
- exploring the relationship between a categorical and a continuous variable.

After the core tasks there is a short, optional section for anyone who wants to go further, followed by two quick questions to help us understand how the session went.

About the data

We will use a data set that is available in R (when you load the `ggplot2` library). The data set is called `diamonds` and it contains the prices and other attributes of almost 54,000 diamonds. The variable details are in Table 1

1. Data exploration
   (a) Create a new R project, and a copy the R markdown file (Independent practice week 1) onto your computer. Load the `ggplot2` library. Make sure you answer the questions using the R markdown. (if you are not able to load the ggplot library download the csv version)
   (b) Check that the data set (`diamonds`) does indeed contain the columns as in Table 1
   (c) Table 1 specifies some ranges for `price, carat, x,y,z, length, table`. Are there any values in the data that are outside the range ? Which ones, and suggest a different range.
   (d) How many levels (or different categories) are in the column labelled `cut`?

variable name	description
price	price in US dollars ($326–$18,823)
carat	weight of the diamond (0.2–5.01)
cut	quality of the cut
color	diamond colour, from D (best) to J (worst)
clarity	a measurement of how clear the diamond is (I1 (worst), SI2, SI1, VS2, VS1, VVS2, VVS1, IF (best))
x	length in mm (0–10.74)
y	width in mm (0–58.9)
z	depth in mm (0–3.18)
depth	total depth percentage $= z/mean(x, y) = 2 * z/(x + y)$(43–79)
table	width of top of diamond relative to widest point (43–95)

Table 1: Columns in the `diamond` data set

(e) Generate a numerical summary for each variable in the data set - take care to use the appropriate technique for each different variable type.

(f) For each variable generate a plot, use different options and make sure the plots are labelled, and suitable to the type of variable.

(g) Explore the graphing of one categorical against one of the continuous variables in this data.

Quick reflection (not marked)

Once you have finished the core tasks above, please take a minute to answer two quick, anonymous questions. This is not marked, is not linked to your grade or profile in any way, and simply helps us understand how the session went and what to focus on next.

1. How far did you get with the core tasks today?
2. What is one thing about this data (or working with it in R) that surprised you or that you found informative and/or that you would like to understand better?

Please answer both questions here: Link to form

Going further (optional)

The tasks below are entirely optional. If you have used R in depth before, or you finish the core tasks with time to spare, try any of these - in any order, none, or all of them. Skipping them has no effect on your progress in this module.

- **Extending (c):** beyond identifying which values fall outside the ranges in Table 1, count how many rows are affected for each variable. Do these look like data-entry errors, or plausible extreme values? What would you check next to decide?
- **Extending (d):** `cut`, `color` and `clarity` all have a natural order (for example `color` runs from D, the best, to J, the worst). Produce frequency and relative frequency tables for `color` and `clarity` as well as `cut`, and look into how to tell R that a categorical variable is *ordered* (see `factor(..., ordered = TRUE)`).
- **Extending (e):** use `aggregate()` to compute the mean (or median) `price` for each level of `cut`. Try grouping by `color` or `clarity` instead - does the pattern surprise you?
- **Extending (f):** `price` and `carat` are both highly skewed. Try plotting one of them on a log scale (in `ggplot2`, `scale_x_log10()` or `scale_y_log10()`) and compare it with the original plot. What difference does this make to what you can see?
- **New - relationships between two continuous variables:** produce a scatter plot of `carat` against `price`, and add a fitted trend line (`geom_smooth()` in `ggplot2`). We have not covered how to formally describe this kind of relationship yet - we will come back to it in Week 5 - but see what you notice in the meantime.